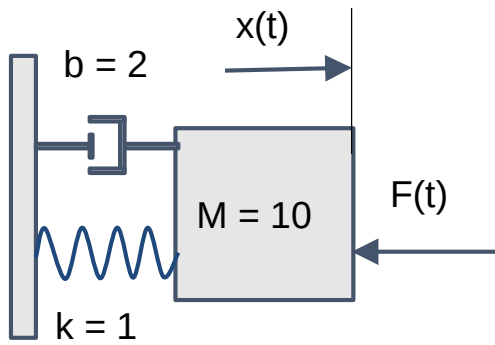




1. Pišemo ODJ za ovaj sistem:

$$Mx'' + bx' + kx = -F(t)$$

$$x(0) = 1, x'(0) = 2$$

2. Određujemo/definišemo šta su ulazi a šta izlazi iz modela:

$$u(t) = F(t) \Rightarrow \text{ulaz u model}$$

$$y_1(t) = x(t), y_2(t) = x'(t) \Rightarrow \text{izlazi iz modela}$$

3. Model treba da formulišemo u obliku:

$$\underline{x}' = A\underline{x} + B\underline{u}$$

$$\underline{y} = C\underline{x} + D\underline{u}$$

4. Kod ovog primera, za izvode promenljivih stanja:

$$x_1 = x, x_2 = x'$$

$$x_1' = x_2$$

$$x_2' = -\frac{k}{M}x_1 - \frac{b}{M}x_2 - \frac{1}{M}F(t)$$

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{M} & -\frac{b}{M} \end{bmatrix}, B = \begin{bmatrix} 0 \\ -\frac{1}{M} \end{bmatrix}$$

5. Za izlaze:

$$\left. \begin{array}{l} y_1 = x(t) = x_1(t) \\ y_2 = x'(t) = x_2(t) \end{array} \right\} \underline{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} F(t)$$

očividno :

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

6. Početni uslovi, jednostavno dobijamo iz smena:

$$\left. \begin{array}{l} x(0) = x_1(0) = 1 \\ x'(0) = x_2(0) = 2 \end{array} \right\} \underline{x}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$